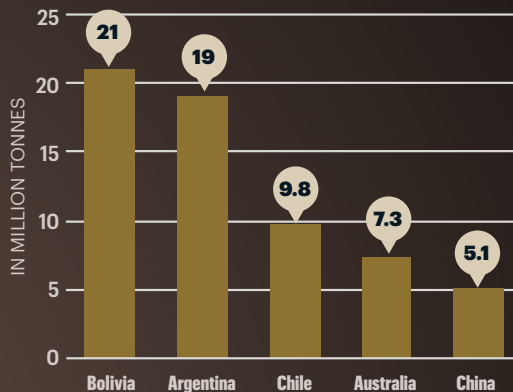


GLOBAL RESERVES

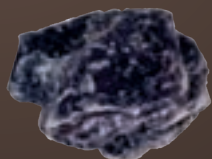
India's lithium reserves, if fully usable, would place it ahead of China in the global pecking order

IDENTIFIED LITHIUM RESOURCES (JANUARY 2022); **SOURCE** UNITED STATES GEOLOGICAL SURVEY



5.9
MILLION
TONNES

Inferred amount of lithium reserves discovered in the Jammu & Kashmir region



METAL MOMENT

Year	2018-19	2019-20	2020-21
Lithium	202	147	173
Lithium-ion	8,574	8,819	8,811

IN ₹ CRORE; VALUE OF TOTAL IMPORTS TO INDIA OF LITHIUM AND LITHIUM-ION BETWEEN 2018-19 AND 2020-21; **SOURCE** MINISTRY OF COMMERCE AND INDUSTRY

that have large lithium reserves (see *Global Reserves*). But the road to getting this domestic treasure into the country's burgeoning EV and green energy spaces will be a long and bumpy one. India will have to drive itself on bigger wheels (build the ecosystem), fix the bad tyre alignment (change policy) and even install new suspensions (provide incentives) to get the ball rolling. Incidentally, the government had already announced a production-linked incentive (PLI) scheme in 2021 to support production of advanced chemistry cell (ACC) energy storage tech in the country. With a financial outlay

of ₹18,100 crore for seven years, the scheme aims to establish a globally competitive ACC-battery manufacturing ecosystem with a capacity of up to 50 GWh.

"This [discovery] can be game-changing depending on how much eventually turns out to be proven and extractable from the ore discovered. The declared discovery of 5.9 million tonnes is very large otherwise and can ensure that domestic needs are addressed," says Anish De, Global Head of Energy and Natural Resources at KPMG. Emphasising the word 'extractable' used by De, industry experts say there are chances of the quantum

of reserves falling by a third when mined. The rationale behind this conservative outlook is simple. The discovery of the reserves in India—that has excited everyone—is in the G3 stage, where the volume, grade and mineral content of the reserves is only estimated. This will be followed by the G2 (general exploration) and G1 (detailed exploration) stages, during which the reserves' actual properties will be confirmed and only after that can commercial mining begin.

Popularly found in different forms of deposits—brine and ore—the extraction of lithium is also different. For the reserves found in the brine reservoirs of South America, salt-rich water is pumped to the surface and into a series of evaporation ponds. As the water evaporates, a slurry of hydrated lime is added to the brine to precipitate the unwanted elements, leaving a brine with an ever-increasing concentration of lithium. Later, this brine is pumped into a recovery facility, where the brine is further purified to remove contaminants and other unwanted elements, followed by chemical treatment, filtration and then treatment with soda ash to collect lithium carbonate precipitate. Finally, this precipitate is washed and dried into the final product. But the lithium discovered in India is ore-based, which has a long and tedious mining process.

After the size, location and quality of the deposits are assessed, firms with experience in mining lithium will be pressed into service to develop the site, which may involve building access roads, constructing buildings and other infrastructure. Setting all this up can take a year or more. Once the basic infra is in place, heavy machinery is deployed to extract the ore, which can be found mixed with sand, stone or other metals.